

Features

- Low Offset Voltage: 5 μ V(max)
- Zero Drift: 0.05 μ V/ $^{\circ}$ C(max)
- No 1/f Noise Corner Down to 0.1Hz
 - Input Noise Voltage 15 nV/ $\sqrt{\text{Hz}}$ @ 1kHz
 - 0.1Hz to 10Hz Noise Voltage 350nVP-P
- Slew Rate: 2.5 V/ μ s
- Bandwidth: 3.5MHz
- Low Supply Current: 500 μ A per Amplifier
- Low Input Bias Current: 50pA Typical
- Rail-to-Rail Output Voltage Range
- High gain, CMRR, PSRR: 130 dB
- 7K HBM ESD Rating
- -40 $^{\circ}$ C to 125 $^{\circ}$ C Operation Range

Description

The family of amplifier are single/dual/quad chopper stabilized zero-drift operational amplifier optimize for single or dual supply operation from 1.8V to 5.5V and ± 0.9 V to ± 2.75 V. TP555X features very low input offset voltage and low noise with no 1/f noise corner down to 0.1Hz. The TP555X is designed to have ultra low offset voltage and offset temperature drift, wide gain bandwidth and rail-to-rail input/output swing while minimizing power consumption.

This family amp TP555X can provide very low offset voltage(max 5 μ V) and near-zero drift over time and temperature with excellent CMRR and PSRR.

The TP5551(single version) is available in SOT23-5, SC70-5 and SO-8 packages. The TP5552(dual version) is offered in MOSP-8, SO-8 package. The TP5554(quad version) is available in TSSOP-14,SO-14 package. All versions are specified for operation from -40 $^{\circ}$ C to 125 $^{\circ}$ C.

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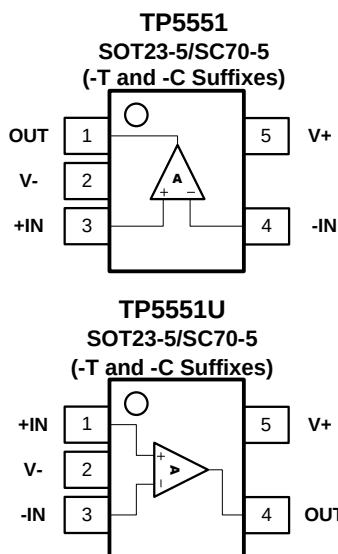
Applications

- Medical Instrumentation
- Temperature Measurements
- Precision current sensing
- Precision Low Drift, Low Frequency ADC Drivers
- Process Control Systems
- Precision Voltage Reference Buffers

Related Zero-Drift RRIO Op-amps

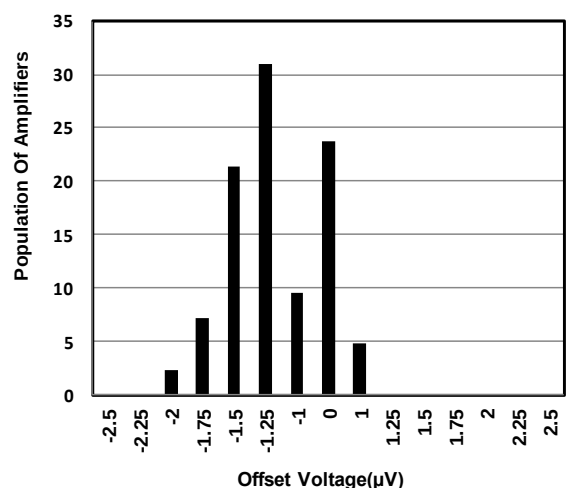
Single	Vos(Max)	Supply Current	e_n @1kHz (nV/ $\sqrt{\text{Hz}}$)	GBW
TP555X	5 μ V	560 μ A	12	3.5 MHz
TP553X	10 μ V	34 μ A	55	350kHz

Pin Configuration (Top View)



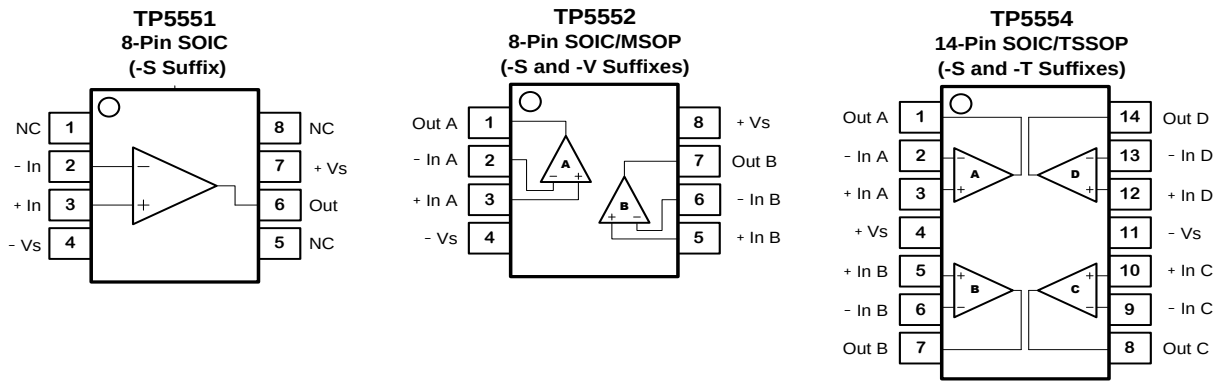
Pin Configuration (Top View)

Offset Voltage Distribution



TP5551/TP5552/TP5554

Ultra Low Noise, 3.5MHz, Zero Drift, RRIO Op-amps



Order Information

Model Name	Order Number	Package	MSL	Transport Media, Quantity	Marking Information
TP5551	TP5551-TR	SOT23-5	3	Tape and Reel, 3000	E51T
	TP5551L1-TR	SOT23-5	1	Tape and Reel, 3000	E51T
	TP5551-CR	SC70-5	3	Tape and Reel, 3000	51C
	TP5551-SR	SO-8	3	Tape and Reel, 4000	TP5551
TP5551U	TP5551U-TR	SOT23-5	3	Tape and Reel, 3000	E51U
	TP5551U-CR	SC70-5	3	Tape and Reel, 3000	51V
TP5552	TP5552-SR	SO-8	3	Tape and Reel, 4000	TP5552
	TP5552L1-SR	SO-8	1	Tape and Reel, 4000	TP5552
	TP5552-VR	MSOP-8	3	Tape and Reel, 3000	TP5552
TP5554	TP5554-SR	SO-14	3	Tape and Reel, 2500	TP5554
	TP5554-TR	TSSOP-14	3	Tape and Reel, 3000	TP5554

Absolute Maximum Ratings Note 1

Supply Voltage:7V
Input Voltage: $V^- - 0.3$ to $V^+ + 0.3$
Input Current: +IN, -IN Note 2 $\pm 20\text{mA}$
Output Current: OUT $\pm 60\text{mA}$
Output Short-Circuit Duration Note 3 Indefinite

Current at Supply Pins $\pm 50\text{mA}$
Operating Temperature Range -40°C to 125°C
Maximum Junction Temperature 150°C
Storage Temperature Range -65°C to 150°C
Lead Temperature (Soldering, 10 sec) 260°C

Note 1: Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. Exposure to any Absolute Maximum Rating condition for extended periods may affect device reliability and lifetime.

Note 2: The inputs are protected by ESD protection diodes to each power supply. If the input extends more than 500mV beyond the power supply, the input current should be limited to less than 10mA.

Note 3: A heat sink may be required to keep the junction temperature below the absolute maximum. This depends on the power supply voltage and how many amplifiers are shorted. Thermal resistance varies with the amount of PC board metal connected to the package. The specified values are for short traces connected to the leads.

ESD, Electrostatic Discharge Protection

Symbol	Parameter	Condition	Minimum Level	Unit
HBM	Human Body Model ESD	ANSI/ESDA/JEDEC JS-001	7	kV
CDM	Charged Device Model ESD	ANSI/ESDA/JEDEC JS-002	2	kV

Ultra Low Noise, 3.5MHz, Zero Drift, RRIO Op-amps

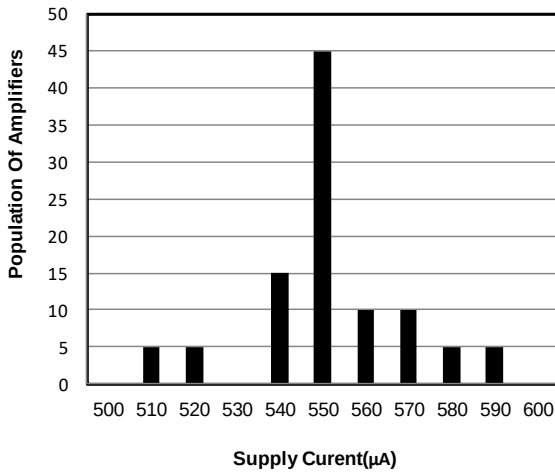
Electrical Characteristics

At $T_A = 27^\circ\text{C}$, $V_{DD}=5\text{V}$, $R_L=10\text{k}\Omega$, $V_{cm}=V_{DD}/2$, unless otherwise noted.

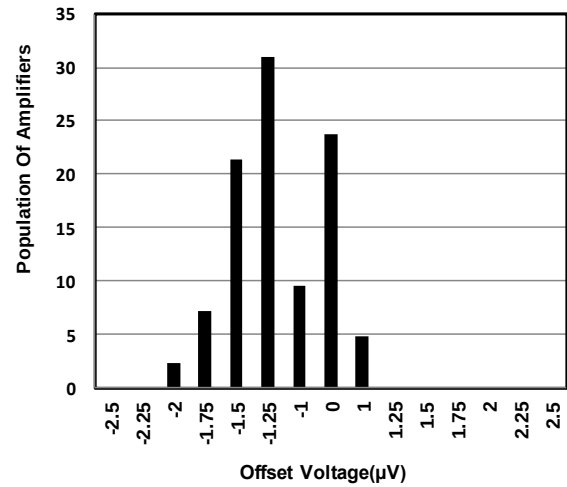
VDD=5V, SYMBOL	PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
V_{DD}	Supply Voltage Range		1.8		5.5	V
I_Q	Quiescent current per amplifier	TP5551		1200	1400	μA
		TP5551, $T_A = -40^\circ\text{C}$ to 85°C			1600	μA
		TP5551, $T_A = -40^\circ\text{C}$ to 125°C			1750	μA
		TP5552/4		550	950	μA
		TP5552/4, $T_A = -40^\circ\text{C}$ to 85°C			1150	μA
		TP5552/4, $T_A = -40^\circ\text{C}$ to 125°C			1300	μA
V_{OS}	input Offset Voltage			± 1	± 5	μV
		$T_A = -40^\circ\text{C}$ to 125°C			± 10	μV
dV_{OS}/dT	vs temperature			0.008	0.05	$\mu\text{V}/^\circ\text{C}$
PSRR	vs power supply	$V_S = +1.8\text{V}$ to $+5.5\text{V}$	110	130		dB
		$V_S = +1.8\text{V}$ to $+5.5\text{V}$, $T_A = -40^\circ\text{C}$ to 125°C	105			
$V_N(p-p)$	input voltage noise, $f=0.01\text{Hz}$ to 1Hz			0.1		μV_{pp}
	input voltage noise, $f=0.1\text{Hz}$ to 10Hz			0.35		μV_{pp}
V_N	Input voltage noise density, $f=1\text{kHz}$			15		$\text{nV}/\sqrt{\text{Hz}}$
C_{IN}	Input capacitor Differential			3		pF
	Input capacitor Common-Mode			2		pF
I_B	Input Current			± 50		pA
	Over temperature			± 200		pA
I_{OS}	Input offset current			± 100		pA
V_{CM}	Common-mode voltage range		(V-)-0.1		(V+)+0.1	V
CMRR	Common-mode rejection ratio	$V_S=5\text{V}$, $V_{CM} = 0.5\text{V}$ to 4.5V	110	130		dB
CMRR	Common-mode rejection ratio	$V_S=5\text{V}$, $V_{CM} = 0\text{V}$ to 5V	100	120		dB
		$V_S=5\text{V}$, $V_{CM} = 0\text{V}$ to 5V , $T_A = -40^\circ\text{C}$ to 125°C	90			dB
V_O	Output Voltage Swing from rail	$R_L=10\text{k}\Omega$		10	25	mV
		$R_L=10\text{k}\Omega$, $T_A = -40^\circ\text{C}$ to 125°C			30	mV
I_{SC}	Short-circuit current			± 50		mA
GBW	Unity Gain Bandwidth	$C_L=100\text{pF}$		3.5		MHz
SR	Slew rate	$G=+1$, $C_L=100\text{pF}$		2.5		V/ μs
t_{or}	Overload recovery time	$G=-10$		35		μs
t_s	Settling time to 0.01%	$C_L=100\text{pF}$		20		μs
A_{VO}	Open-Loop Voltage Gain	(V-)+100mV< V_O <(V+)-100mV, $R_L = 100\text{k}\Omega$	100	120		dB
		(V-)+100mV< V_O <(V+)-100mV, $R_L = 100\text{k}\Omega$, $T_A = -40^\circ\text{C}$ to 125°C	90			dB
θ_{JA}	Thermal Resistance Junction to Ambient	SOT23-5		200		$^\circ\text{C}/\text{W}$
		MSOP-8		210		
		SO-8		158		
		SC70-5		250		
		SO-14		83		
		TSSOP-14		100		

Typical Performance Characteristics

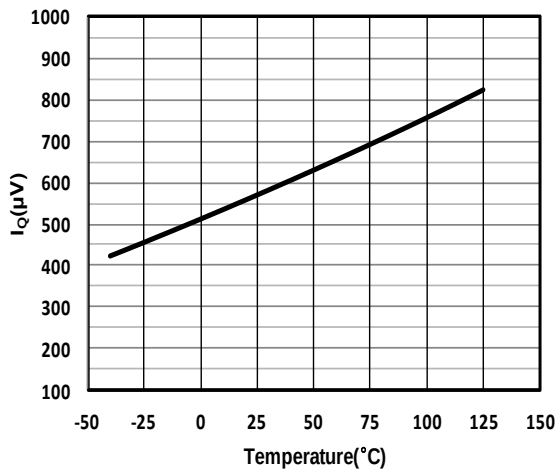
Supply Current Distribution



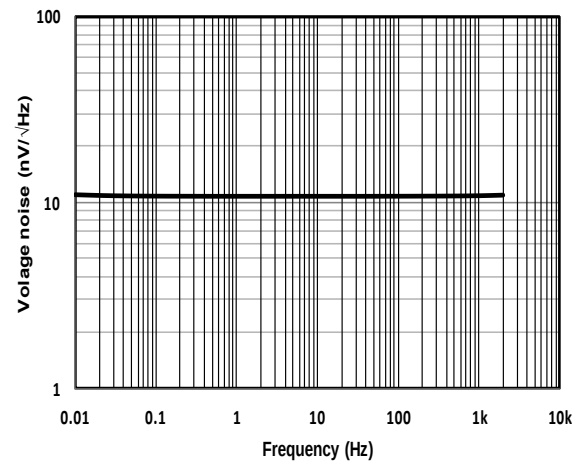
Offset Voltage Distribution



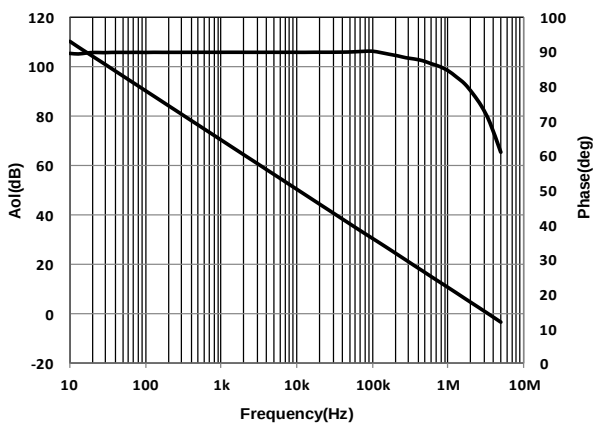
Quiescent Current vs Temperature



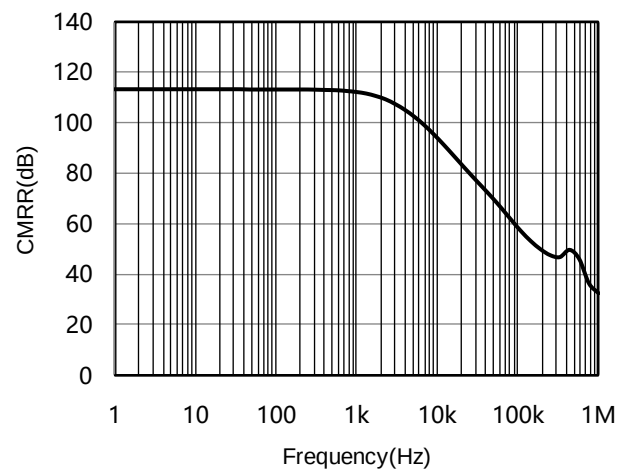
Voltage Noise Spectral Density vs Frequency



OPEN-LOOP GAIN vs FREQUENCY

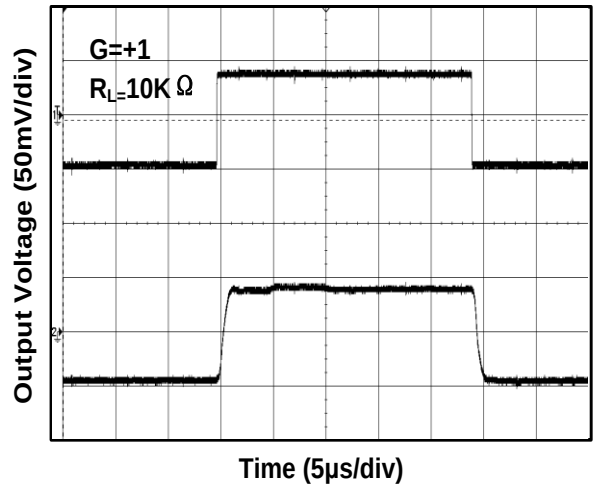


CMRR vs FREQUENCY

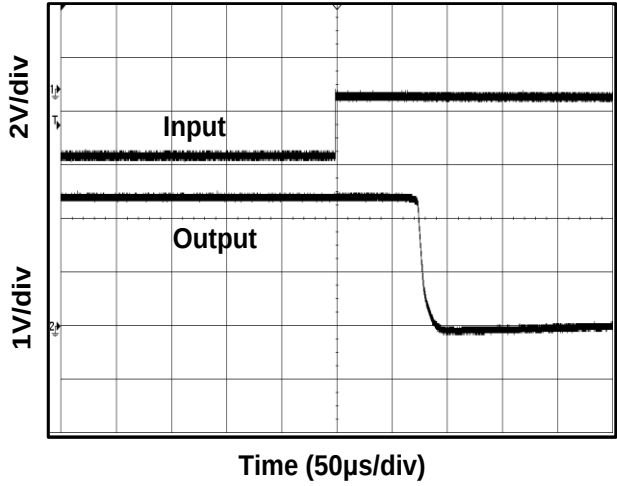


Typical Performance Characteristics (continue)

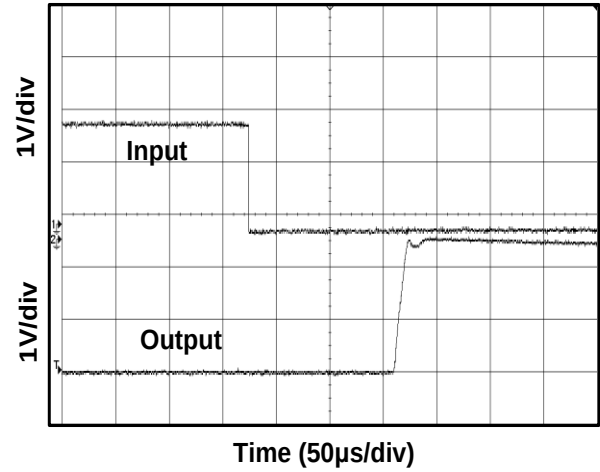
Small-Scale Step Response



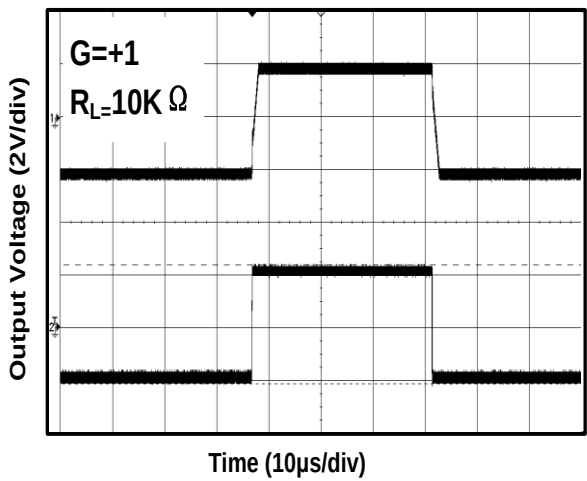
Positive Over-Voltage Recovery



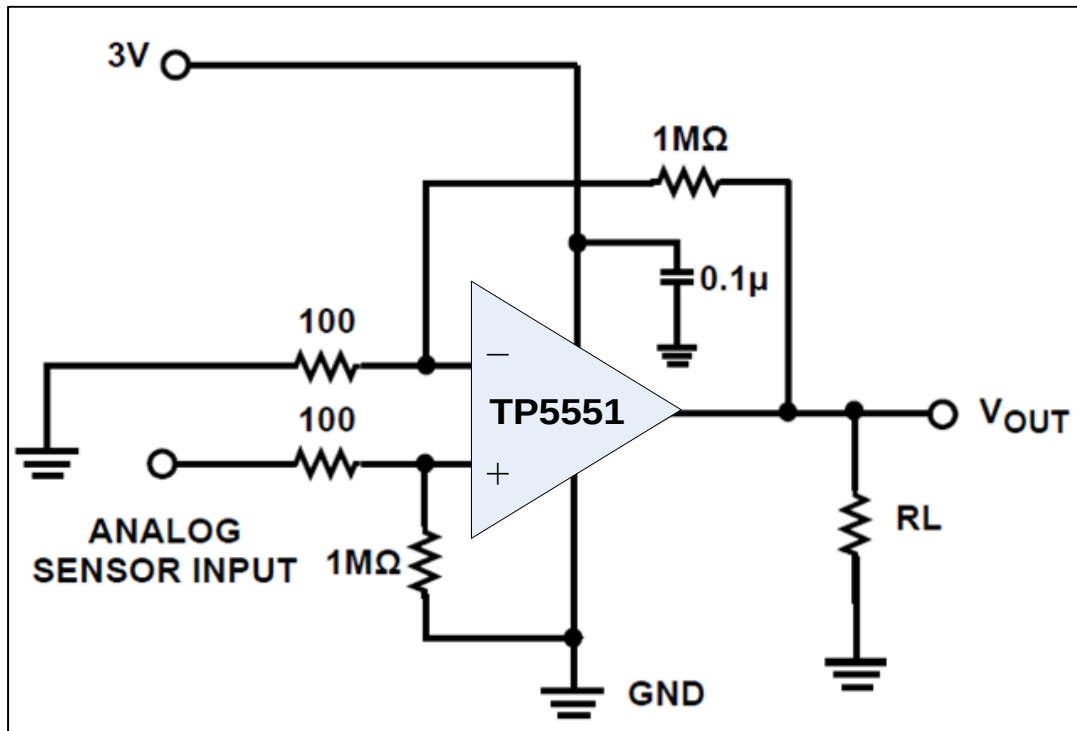
Negative Over-Voltage Recovery



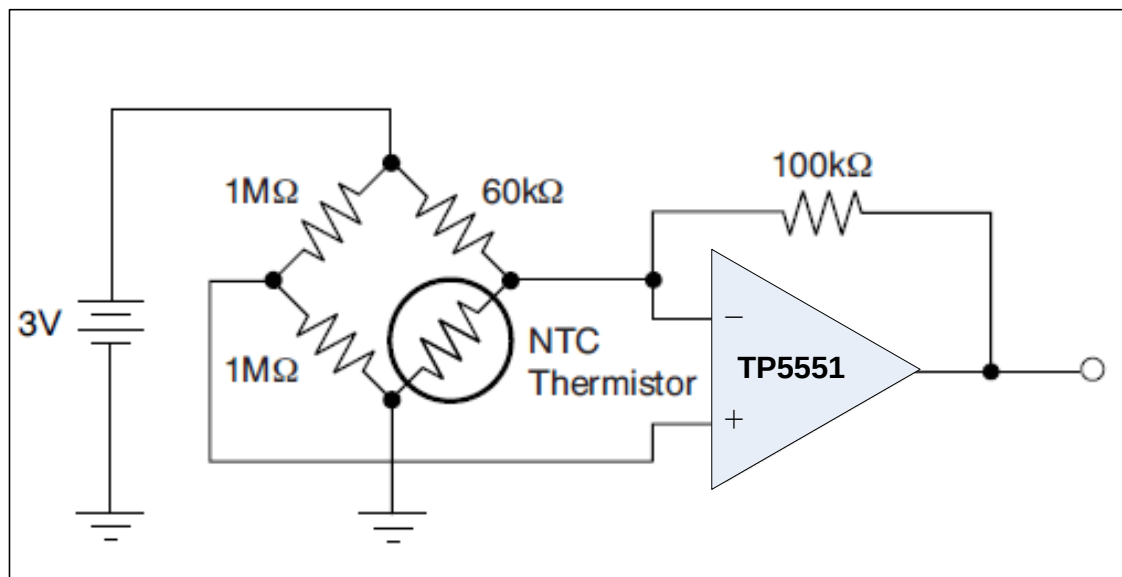
Large-Scale Step Response



TYPICAL APPLICATIONS



Single Supply, High Gain Amplifier, $A_V = 10,000$ V/V



Thermistor Measurement

Ultra Low Noise, 3.5MHz, Zero Drift, RRIO Op-amps

Pin Functions

-IN: Inverting Input of the Amplifier.

+IN: Non-Inverting Input of Amplifier.

OUT: Amplifier Output. The voltage range extends to within mV of each supply rail.

V+ or +V_s: Positive Power Supply. Typically the voltage is from 1.8V to 5.5V. Split supplies are possible as long as the voltage between V+ and V- is between 1.8V and

5.5V. A bypass capacitor of 0.1μF as close to the part as possible should be used between power supply pins or between supply pins and ground.

V- or -V_s: Negative Power Supply. It is normally tied to ground. It can also be tied to a voltage other than ground as long as the voltage between V+ and V- is from 1.8V to 5.5V. If it is not connected to ground, bypass it with a capacitor of 0.1μF as close to the part as possible.

Operation

The TP5551/2/4 op amps are zero drift, rail-to-rail operation amplifiers that can be run from a single-supply voltage. They use an auto-calibration technique with a time-continuous 3.5MHz op amp in the signal path while consuming only 550μA of supply current per channel. This amplifier is zero-corrected with an 150kHz clock. Upon power-up, the amplifier requires approximately 100μs to achieve specified V_{os} accuracy. This design has no aliasing or flicker noise.

Applications Information

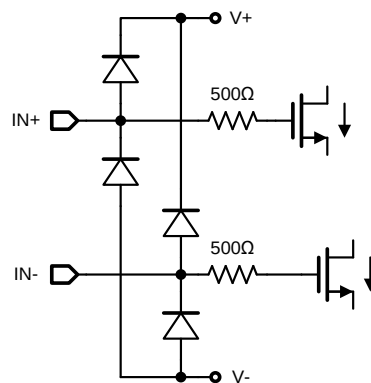
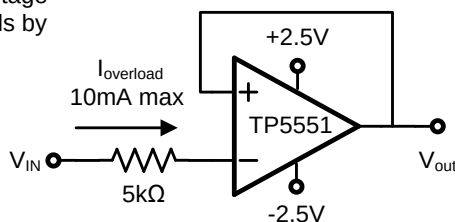
Rail-To-Rail Input And Output

The TP5551/2/4 feature rail-to-rail input and output with a supply voltage from 1.8V to 5.5 V. This allows the amplifier inputs to have a wide common mode range(50mV beyond supply rails)while maintaining high CMRR(120dB) and maximizes the signal to noise ratio of the amplifier by having the V_{OH} and V_{OL} levels be at the V+ and V- rails, respectively.

Input Protection

The TP5551/2/4 have internal ESD protection diodes that are connect between the inputs and supply rail. When either input exceeds one of the supply rails by more than 300mV, the ESD diodes become forward biased and large amounts of current begin to flow through them. Without current limiting, this excessive fault current causes permanent damage to the device. Thus an external series resistor must be used to ensure the input currents never exceed 10mA (see Figure xx).

Current-limiting resistor required if input voltage exceeds supply rails by >0.5V.



INPUT ESD DIODE CURRENT LIMITING- UNITY GAIN

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Low Input Referred Noise

Flicker noise, as known as 1/f noise, is inherent in semiconductor devices and increases as frequency decreases. So at lower frequencies, flicker noise dominates, causing higher degrees of error for sub-Hertz frequencies or dc precision application.

The TP5551/2/4 amplifiers are chopper stabilized amplifiers, the flicker noise is reduced greatly because of this technique. This reduction in 1/f noise allows the TP5551/2/4 to have much lower noise at dc and low frequency compared to standard low noise amplifier.

Residual voltage ripple

The chopping technique can be used in amplifier design due to the internal notch filter. Although the chopping related voltage ripple is suppressed, higher noise spectrum exists at the chopping frequency and its harmonics due to residual ripple.

So if the frequency of input signal is nearby the chopping frequency, the signal maybe interfered by the residue ripple. To further suppress the noise at the chopping frequency, it is recommended that a post filter be placed at the output of the amplifier.

Broad Band And External Resistor Noise Considerations

The total broadband noise output from any amplifier is primarily a function of three types of noise: input voltage noise from the amplifier, input current noise from the amplifier, and thermal (Johnson) noise from the external resistors used around the amplifier. These noise sources are not correlated with each other and their combined noise can be summed in a root sum squared manner. The full equation is given as:

$$e_n \text{ total} = [e_n^2 + 4kTR_s + (i_n \times R_s)^2]^{1/2}$$

Where:

e_n = the input voltage noise density of the amplifier.

i_n = the input current noise of the amplifier.

R_s = source resistance connected to the noninverting terminal.

K = Boltzmann 's constant ($1.38 \times 10^{-23} \text{J/K}$). T = ambient temperature in Kelvin (K).

The total equivalent rms noise over a specific bandwidth is expressed as:

$$e_{n,rms} = e_n \text{ total} \times \sqrt{BW}$$

The input voltage noise density (e_n) of the TP555x is 55 nV/√Hz, and the input current noise can be neglected. When the source resistance is 190 kΩ, the voltage noise contribution from the source resistor and the amplifier are equal. With source resistance greater than 190 kΩ, the overall noise of the system is dominated by the Johnson noise of the resistor itself.

High Source Impedance Application

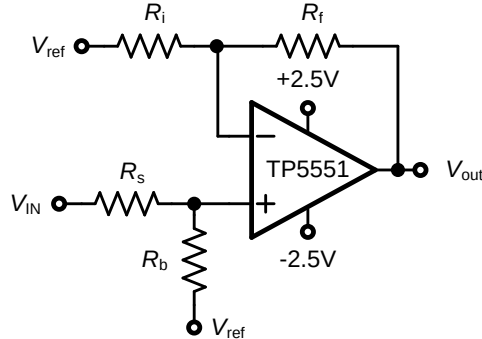
The TP5551/2/4 uses switches at the chopper amplifier input, the input signal is chopped at 125kHz to reduce input offset voltage down to 10μV. The dynamic behavior of these switches induces a charge injection current to the input terminals of the amplifier. The charge injection current has a DC path to ground through the resistances seen at the input terminals of the amplifier. Higher input impedance cause an apparent shift in the input bias current of the amplifier.

Because the chopper amplifier has charge injection currents at each terminal, the input offset current will be larger than standard amplifiers. The I_{os} of TP5551/2/4 are 150pA under the typical condition. So the input impedance should be balanced across each input(see Figure xx). The input impedance of the amplifier should be matched between the IN+ and IN- terminals to minimize total input offset current. Input offset currents show up as an additional output offset voltage, as shown in the following equation:

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$$V_{os,total} = V_{os} - R_f \times I_{os}$$

For a gain configure using 1MΩ feedback resistor, a 150pA total input offset current will have an additional output offset voltage of 0.15mV. By keeping the input impedance low and balanced across the amplifier inputs, the input offset current effect will be suppress efficiently.



Figxx Circuit Implication for reducing Input offset current effect

PCB Surface Leakage

In applications where low input bias current is critical, Printed Circuit Board (PCB) surface leakage effects need to be considered. Surface leakage is caused by humidity, dust or other contamination on the board. It is recommended to use multi-layer PCB layout and route the OPA's -IN and +IN signal under the PCB surface.

The effective way to reduce surface leakage is to use a guard ring around sensitive pins (or traces). The guard ring is biased at the same voltage as the sensitive pin. An example of this type of layout is shown in Figure 2 for Inverting Gain application.

1. For Non-Inverting Gain and Unity-Gain Buffer:

- Connect the non-inverting pin (V_{IN+}) to the input with a wire that does not touch the PCB surface.
- Connect the guard ring to the inverting input pin (V_{IN-}). This biases the guard ring to the Common Mode input voltage.

2. For Inverting Gain and Trans-impedance Gain Amplifiers (convert current to voltage, such as photo detectors):

- Connect the guard ring to the non-inverting input pin (V_{IN+}). This biases the guard ring to the same reference voltage as the op-amp (e.g., $V_{DD}/2$ or ground).
- Connect the inverting pin (V_{IN-}) to the input with a wire that does not touch the PCB surface.

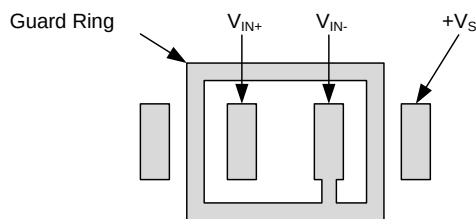


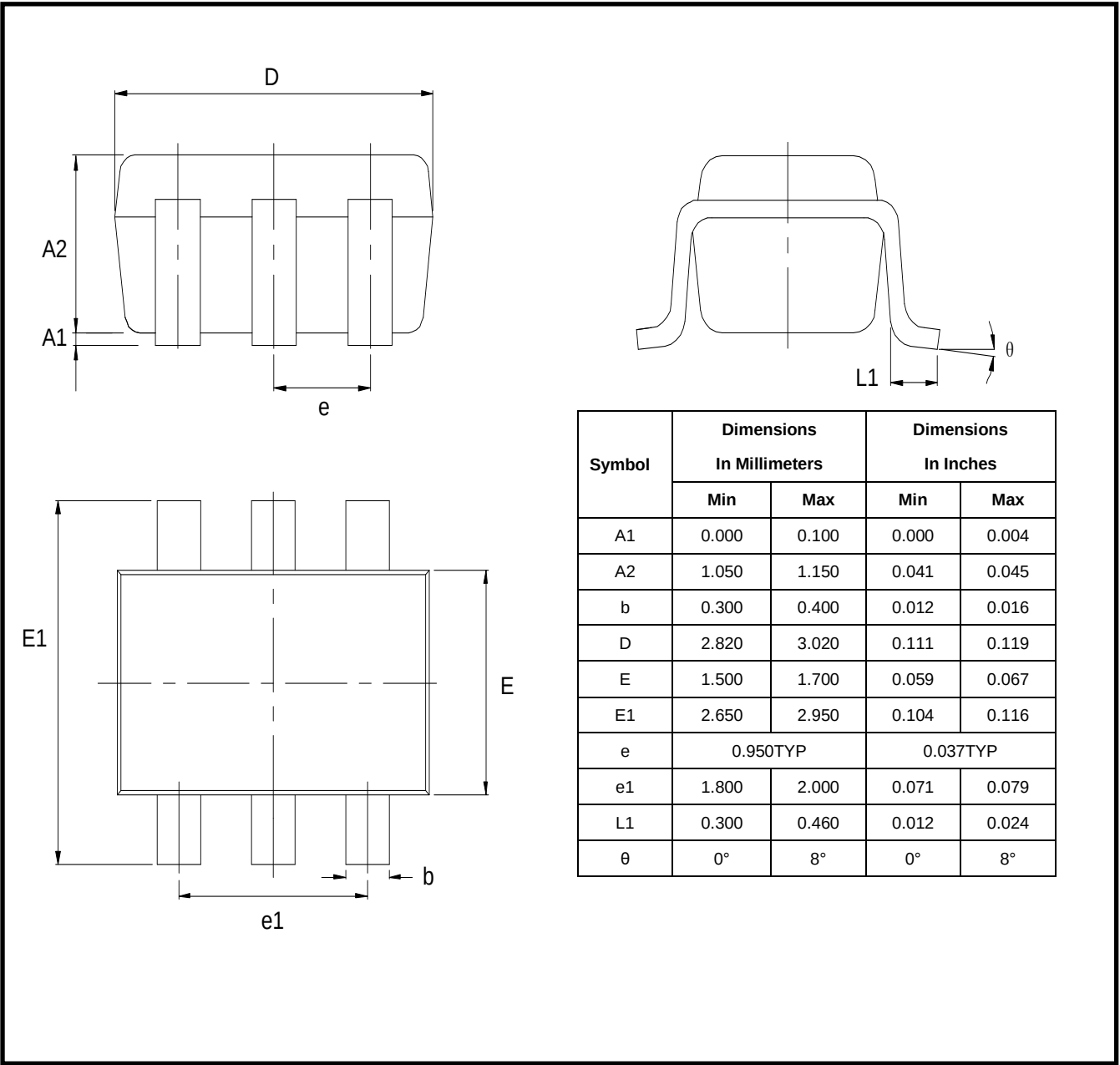
Figure The Layout of Guard Ring

TP5551/TP5552/TP5554

Ultra Low Noise, 3.5MHz, Zero Drift, RRIO Op-amps

Package Outline Dimensions

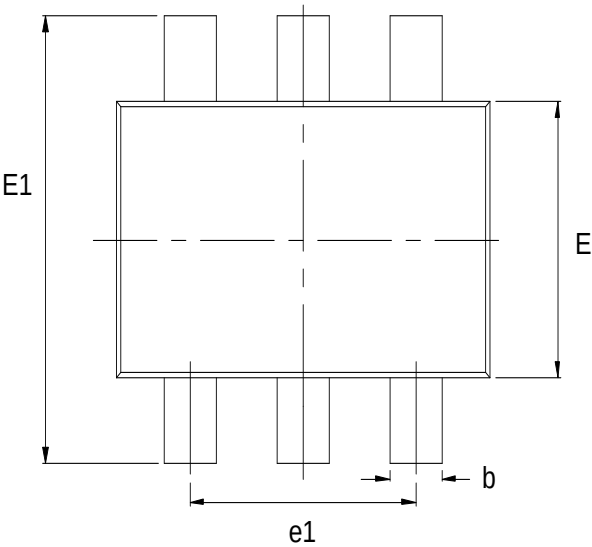
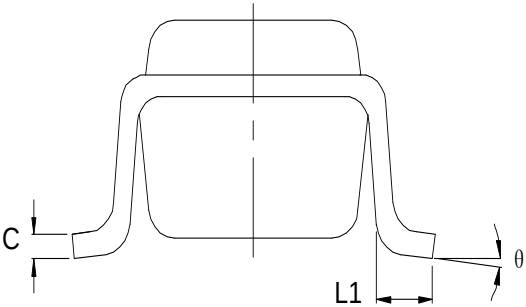
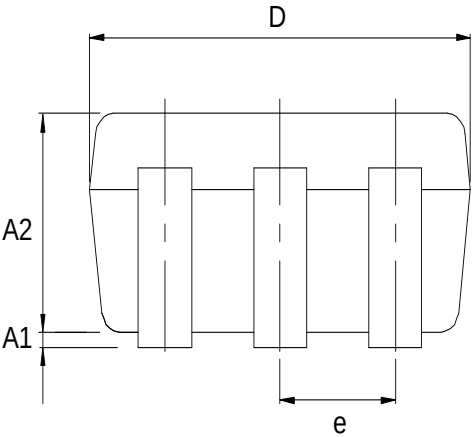
SOT23-5 / SOT23-6



Ultra Low Noise, 3.5MHz, Zero Drift, RRIO Op-amps

Package Outline Dimensions

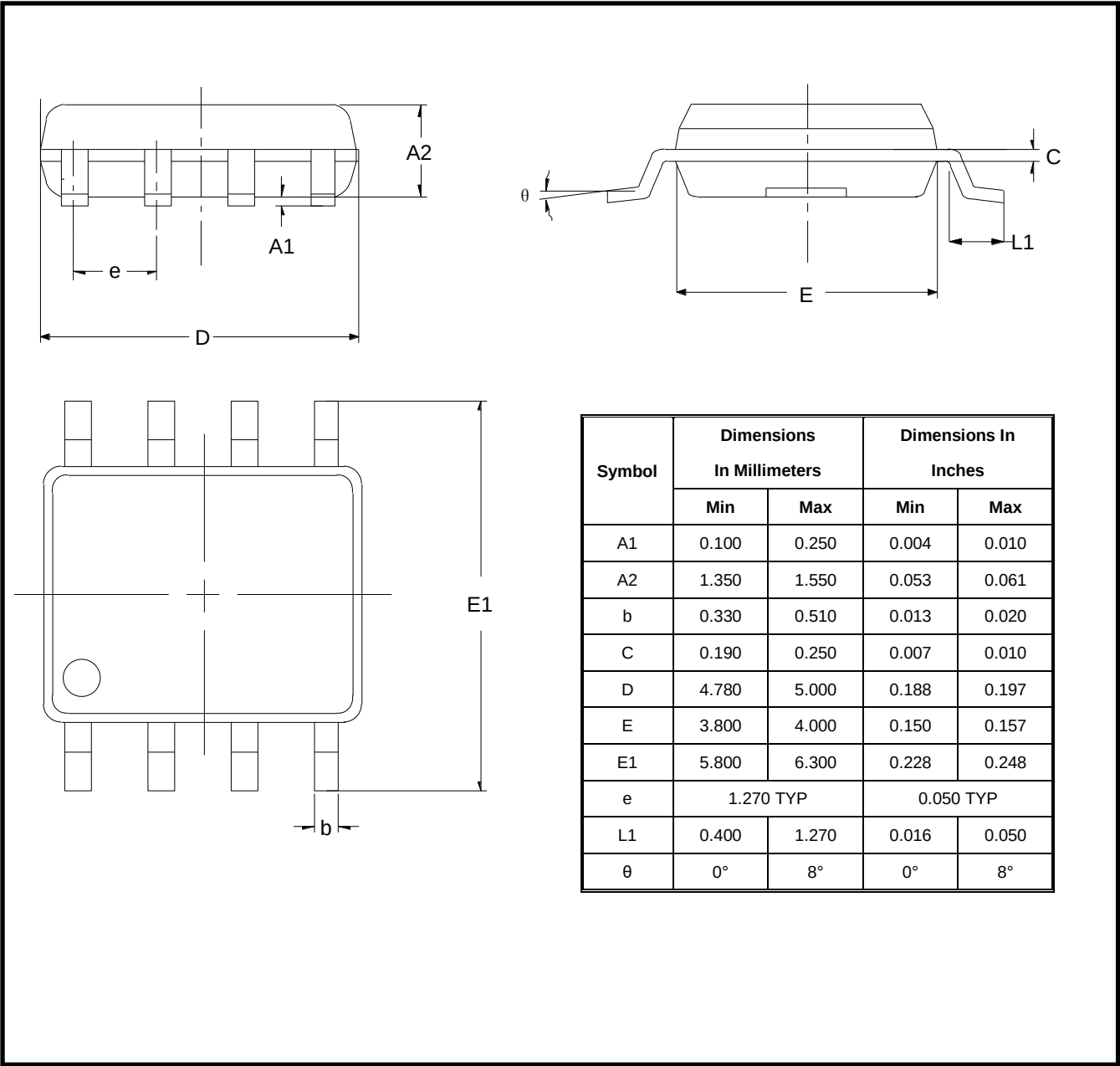
SC-70-6 (SOT363)



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A1	0.000	0.100	0.000	0.004
A2	0.900	1.000	0.035	0.039
b	0.150	0.350	0.006	0.014
C	0.080	0.150	0.003	0.006
D	2.000	2.200	0.079	0.087
E	1.150	1.350	0.045	0.053
E1	2.150	2.450	0.085	0.096
e	0.650TYP		0.026TYP	
e1	1.200	1.400	0.047	0.055
L1	0.260	0.460	0.010	0.018
theta	0°	8°	0°	8°

Package Outline Dimensions

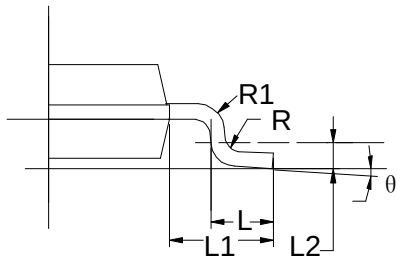
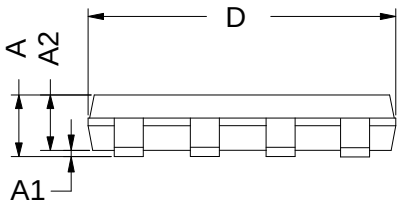
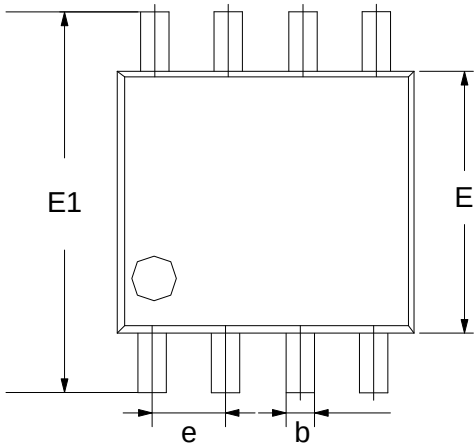
SO-8 (SOIC-8)



Ultra Low Noise, 3.5MHz, Zero Drift, RRIO Op-amps

Package Outline Dimensions

MSOP-8



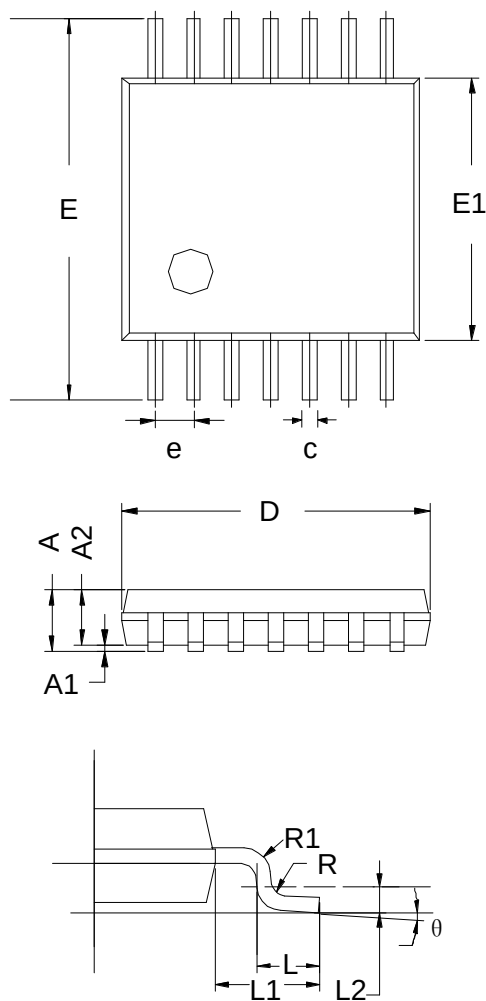
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	0.800	1.200	0.031	0.047
A1	0.000	0.200	0.000	0.008
A2	0.760	0.970	0.030	0.038
b	0.30 TYP		0.012 TYP	
C	0.15 TYP		0.006 TYP	
D	2.900	3.100	0.114	0.122
e	0.65 TYP		0.026	
E	2.900	3.100	0.114	0.122
E1	4.700	5.100	0.185	0.201
L1	0.410	0.650	0.016	0.026
θ	0°	6°	0°	6°

TP5551/TP5552/TP5554

Ultra Low Noise, 3.5MHz, Zero Drift, RRIO Op-amps

Package Outline Dimensions

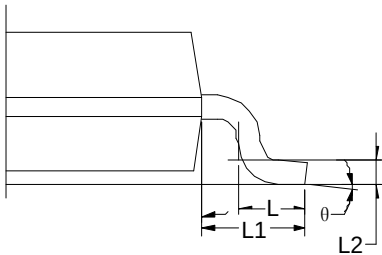
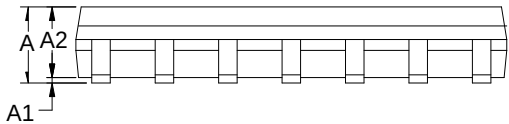
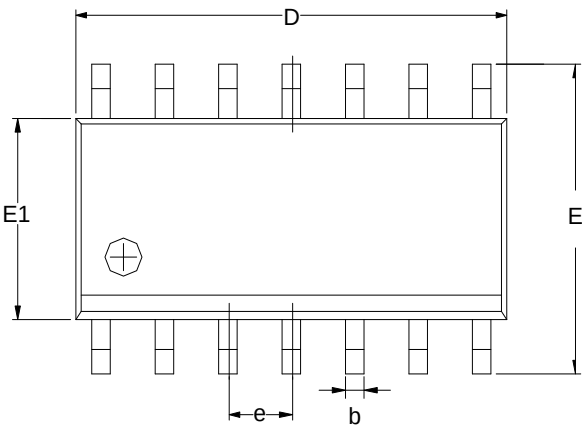
TSSPO-14



Symbol	Dimensions In Millimeters		
	MIN	TYP	MAX
A	-	-	1.20
A1	0.05	-	0.15
A2	0.90	1.00	1.05
b	0.20	-	0.28
c	0.10	-	0.19
D	4.86	4.96	5.06
E	6.20	6.40	6.60
E1	4.30	4.40	4.50
e	0.65 BSC		
L	0.45	0.60	0.75
L1	1.00 REF		
L2	0.25 BSC		
R	0.09	-	-
θ	0°	-	8°

Package Outline Dimensions

SO-14 (SOIC-14)



Symbol	Dimensions In Millimeters		
	MIN	TYP	MAX
A	1.35	1.60	1.75
A1	0.10	0.15	0.25
A2	1.25	1.45	1.65
b	0.36		0.49
D	8.53	8.63	8.73
E	5.80	6.00	6.20
E1	3.80	3.90	4.00
e	1.27 BSC		
L	0.45	0.60	0.80
L1	1.04 REF		
L2	0.25 BSC		
θ	0°		8°

TP5551 / TP5552/TP5554

Ultra Low Noise, 3.5MHz, Zero Drift, RRIO Op-amps

Revision History

2018/8/30	Rev B	Update Full Temperature Specification
2020/5/25	Rev B.1	Update Specification of PSRR: 25degree: min 95-> min 110; -40 to 125degree: min 85->min 105